CSE2260 - Principles of Programming Languages
PROJECT II (Due 25/05/2026 23:59)

In this project, you will implement the software described below using **Python** programming language.

A family tree warehouse application (FTWA): creates and manages family trees. FTWA keeps a track of the ancestral hierarchy of the user's family.

There exist several rules that each group should implement:

- Users should be able to add new members to the family tree and update people in family tree.
- Maintaining only one family tree is enough.
- Each person should keep at least the following information: gender [male, female], name [string], surname [string], birth date [date], death date [date], father [one person], mother [one person], children [zero or more child]. A class named Person should be created.
- Users should be able add any person as child. Any other relationship is prohibited while entering. So, you should be able to calculate relationships when asked by the user.
- Users should be able to reach the following information: alive or dead, age, the level in the family tree. Your program will calculate this information when asked by the user.
- Users should be able to ask the relationship between 2 people.
- The application should display the tree as meaningful diagram. **The menu, menu options, input and output formats and their order must be exactly the same as shown in the visuals and must not be changed.**
- The application should display the relationship types in Turkish. The following types should be supported.
    - Anne, Baba, Oğul, Kız, Erkek Kardeş, Kız Kardeş, Abla, Abi
    - Büyükanne, Büyükbaba, Torun
    - Amca, Hala, Dayı, Teyze
    - Yeğen, Kuzen
    - Enişte, Yenge
    - Kayınvalide, Kayınpeder, Gelin, Damat
    - Bacanak, Baldız, Elti, Kayınbirader
- Anyone can marry with anyone in the family tree except father, mother, sibling, uncle, aunt and grandparents. If a marriage violates this rule, your program should warn the user and reject the marriage.

- Birth and death dates should be realistic. For instance, a person cannot have a child after he/she dies. The application should not allow non-realistic birth and death dates and should warn the user.

- The application should give warnings when any of the spouse is below 18 years old. The program should warn the user and reject the marriage if any of the spouse is below 18 years old.

- The main menu should continuously prompt the user for commands until the user selects the logout option.

- For each specified kinship relationship (uncle, cousin, older brother, son, etc.), a function should be written that includes the name of that kinship. The function should take two people as parameters and determine if the relevant kinship exists between these two people.

- When option number 1 is selected in the main menu, the `*find_relation*` function is called, and if a kinship relationship exists between the specified individuals, its type is printed to the screen; otherwise, "No Kinship Relationship" is printed. This function uses the functions written for kinship relationships mentioned in the previous section.

- To add children to the family tree, a function called `*add_child*` will be written, and this function will be called if the option number 2 is selected in the main menu and the sub-option number 1 is selected.

- If the family tree is empty, the user is prompted to add the root parents before adding children. The mother and father information of the root parents is added as "none".

- When the Update Person option is selected, the function named '*update_person*' should be called to perform the necessary update. This function will be called if the option number 2 is selected in the main menu and the sub-option number 2 is selected.

- When a child is added to the family tree, the number of children of both the mother and father must increase by one. The child's level in the tree is also assigned to the next level, whichever parent is lower in the tree.

- When option 5 is selected from the main menu, the function named `*add_marriage*` should be called to perform the necessary checks and to accept or reject the marriage.

- If a member of a family tree is marrying someone who is not a member, the information of the person they are marrying must be requested.

    - The parents' information for individuals outside the family tree is added as "none".

    - If the marriage is valid, the new person is added to the family tree.

    - Individuals who are not members of the family tree are not shown in the family tree unless they have entered into a valid marriage.

- If a marriage proposal is rejected, simply stating that it is invalid is not enough. The reason must also be stated, such as marriage under the age of 18, or the type of kinship relationship that prevents the marriage, etc.

- If a child cannot be added to the family tree, the reason should be stated. For example, the child's mother or father may not be in the tree, or the child may have been born after the death of their mother or father, etc.

Assumptions:

- You can assume that there will not be two people having both the same name and surname.

- You can assume that update operation will be performed for just the birthdate and death date. You can assume that update operation will not affect the previously successfully completed operations. But the operations after the update will be done according to the updated data.

Submission rules:

1.You should submit the project through google classroom.

2.This project is a group project. You are going to work in groups of **3 people**. Only the group representative will upload the project.

3.You will submit the source code and the report as a single .zip file.The name of the zip file should be studentnumber1_studentnumber2_studentnumber3.zip

4.In the report you should provide the information listed below:

0.**How to compile and run your code.**

1.How you create the tree and the brief explanation of the related code.

2.How you add a new person into the tree and the brief explanation of the related code.

3.How you perform the update operation and the brief explanation of the related code.

4.How you calculate whether a person is alive or dead, his/her age and his/her corresponding level in the family tree and the brief explanation of the related code.

5.How you find the relationships between people and the brief explanation of the related code.

6.How you print the family tree and the brief explanation of the related code.

7.How you control the marriage validity (both for invalid relationship types and being under 18 years old) and the brief explanation of the related code.

8.Complete/incomplete parts and the parts that are not working as supposed to be.

9.Execution screenshots for:

i.adding person into the family tree.

ii.updating person information.

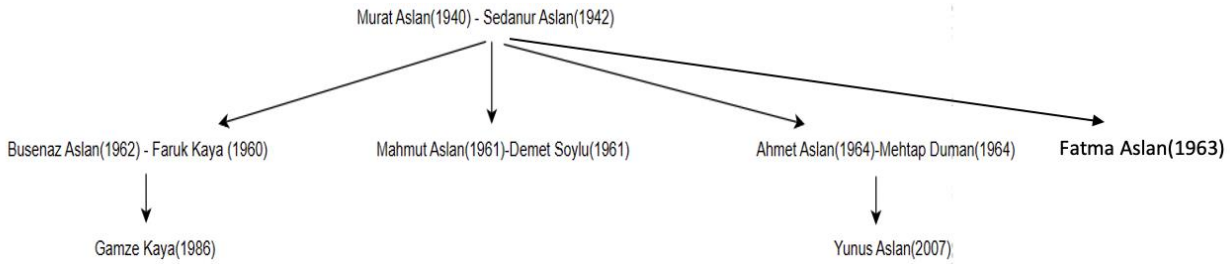
iii.printing the family tree.

iv.printing the information about a person.

v.finding relationships between people.

vi.adding marriage.

Sample Run: You can find the construction and some operations for the tree given below:



Sample screenshots

1. Initially tree is empty. The root members are added and the tree is printed.

```
================================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
================================================
Please choose an operation: 4
Empty Family Tree.

================================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
================================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 1

Empty Family tree. Adding root members

--- Father ---
Enter Name and Surname: Murat Aslan
Enter Birthdate: 1940
Murat Aslan added.

--- Mother ---
Enter Name and Surname: Sedanur Aslan
Enter Birthdate: 1942
Sedanur Aslan added.
success: Murat Aslan and Sedanur Aslan married.

================================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
================================================
Please choose an operation: 4
---LEVEL 0---
Murat Aslan-Sedanur Aslan
```

2. how to add a child

```
==============================================
Please choose an operation: 3
Enter Name and Surname: Sedanur Aslan
Age: 84
Level = 0
Total Children: 0
Status: Alive.

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 1

--- Adding new family member ---
Please type the father name and surname: Murat Aslan
please type the mother name and surname: Sedanur Aslan
Enter Child Name and Surname: Busenaz Aslan
Enter Birthdate: 1962
Enter Death date ('none' if alive): none
Enter Gender (m/f): f
Busenaz Aslan added.
==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 4
---LEVEL 0---
Murat Aslan-Sedanur Aslan
---LEVEL 1---
Busenaz Aslan

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 3
Enter Name and Surname: Murat Aslan
Age: 86
Level = 0
Total Children: 1
Status: Alive.

==============================================
```

```
==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 1

--- Adding new family member ---
Please type the father name and surname: Murat Aslan
please type the mother name and surname: Sedanur Aslan
Enter Child Name and Surname: Mahmut Aslan
Enter Birthdate: 1961
Enter Death date ('none' if alive): none
Enter Gender (m/f): m
Mahmut Aslan added.

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 4
---LEVEL 0---
Murat Aslan-Sedanur Aslan
---LEVEL 1---
Busenaz Aslan
Mahmut Aslan

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 3
Enter Name and Surname: Sedanur Aslan
Age: 84
Level = 0
Total Children: 2
Status: Alive.

==============================================
```

3. Valid Marriage

```
==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 5
Enter Name and Surname of the first person: Faruk Kaya
Enter Name and Surname of the second person: Busenaz Aslan

'Faruk Kaya' not found in the tree. Asking details to add as a new spouse...
Gender of Faruk Kaya (m/f): m
Birth year of Faruk Kaya: 1960
Death year of Faruk Kaya ('none' if alive): none
Faruk Kaya added to the tree as a new spouse.
success: Faruk Kaya and Busenaz Aslan married.

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 4
---LEVEL 0---
Murat Aslan-Sedanur Aslan
---LEVEL 1---
Busenaz Aslan-Faruk Kaya
Mahmut Aslan
```

```
==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 1

--- Adding new family member ---
Please type the father name and surname: Murat Aslan
please type the mother name and surname: Sedanur Aslan
Enter Child Name and Surname: Ahmet Aslan
Enter Birthdate: 1964
Enter Death date ('none' if alive): none
Enter Gender (m/f): m
Ahmet Aslan added.

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 1

--- Adding new family member ---
Please type the father name and surname: Murat Aslan
please type the mother name and surname: Sedanur Aslan
Enter Child Name and Surname: Fatma Aslan
Enter Birthdate: 1963
Enter Death date ('none' if alive): none
Enter Gender (m/f): f
Fatma Aslan added.
```

```

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 5
Enter Name and Surname of the first person: Mahmut Aslan
Enter Name and Surname of the second person: Demet Soylu

'Demet Soylu' not found in the tree. Asking details to add as a new spouse...
Gender of Demet Soylu (m/f): f
Birth year of Demet Soylu: 1961
Death year of Demet Soylu ('none' if alive): none
Demet Soylu added to the tree as a new spouse.
success: Mahmut Aslan and Demet Soylu married.

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 4
---LEVEL 0---
Murat Aslan-Sedanur Aslan
---LEVEL 1---
Busenaz Aslan-Faruk Kaya
Mahmut Aslan-Demet Soylu
Ahmet Aslan
Fatma Aslan
```

```
============================================
1-) Ask relation        | 2-) Add/Update person
3-) Get info of person  | 4-) Print the family tree
5-) Add marriage        | 6-) Terminate program
============================================
Please choose an operation: 5
Enter Name and Surname of the first person: Ahmet Aslan
Enter Name and Surname of the second person: Mehtap Duman

'Mehtap Duman' not found in the tree. Asking details to add as a new spouse...
Gender of Mehtap Duman (m/f): f
Birth year of Mehtap Duman: 1964
Death year of Mehtap Duman ('none' if alive): none
Mehtap Duman added to the tree as a new spouse.
success: Ahmet Aslan and Mehtap Duman married.

============================================
1-) Ask relation        | 2-) Add/Update person
3-) Get info of person  | 4-) Print the family tree
5-) Add marriage        | 6-) Terminate program
============================================
Please choose an operation: 4
---LEVEL 0---
Murat Aslan-Sedanur Aslan
---LEVEL 1---
Busenaz Aslan-Faruk Kaya
Mahmut Aslan-Demet Soylu
Ahmet Aslan-Mehtap Duman
Fatma Aslan
```

4. Update Person

```
============================================
1-) Ask relation        | 2-) Add/Update person
3-) Get info of person  | 4-) Print the family tree
5-) Add marriage        | 6-) Terminate program
============================================
Please choose an operation: 3
Enter Name and Surname: Faruk Kaya
Age: 66
Level = 1
Total Children: 0
Status: Alive.

============================================
1-) Ask relation        | 2-) Add/Update person
3-) Get info of person  | 4-) Print the family tree
5-) Add marriage        | 6-) Terminate program
============================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 2
1-Update Birth Year / 2-Update Death Year / 0-Cancel: 2
Enter Name and Surname of the person: Faruk Kaya
Enter Year: 2015
Faruk Kaya Year of death updated.

============================================
1-) Ask relation        | 2-) Add/Update person
3-) Get info of person  | 4-) Print the family tree
5-) Add marriage        | 6-) Terminate program
============================================
Please choose an operation: 3
Enter Name and Surname: Faruk Kaya
Age: 55
Level = 1
Total Children: 0
Status: Dead.
```

```
============================================
1-) Ask relation        | 2-) Add/Update person
3-) Get info of person  | 4-) Print the family tree
5-) Add marriage        | 6-) Terminate program
============================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 1

--- Adding new family member ---
Please type the father name and surname: Faruk Kaya
please type the mother name and surname: Busenaz Aslan
Enter Child Name and Surname: Gamze Kaya
Enter Birthdate: 2020
Enter Death date ('none' if alive): none
Enter Gender (m/f): f
Error: Father (Faruk Kaya) died before the birth of the child.

============================================
1-) Ask relation        | 2-) Add/Update person
3-) Get info of person  | 4-) Print the family tree
5-) Add marriage        | 6-) Terminate program
============================================
Please choose an operation: 4
---LEVEL 0---
Murat Aslan-Sedanur Aslan
---LEVEL 1---
Busenaz Aslan-Faruk Kaya
Mahmut Aslan-Demet Soylu
Ahmet Aslan-Mehtap Duman
Fatma Aslan
```

```
============================================
1-) Ask relation        | 2-) Add/Update person
3-) Get info of person  | 4-) Print the family tree
5-) Add marriage        | 6-) Terminate program
============================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 1

--- Adding new family member ---
Please type the father name and surname: Faruk Kaya
please type the mother name and surname: Busenaz Aslan
Enter Child Name and Surname: Gamze Kaya
Enter Birthdate: 1986
Enter Death date ('none' if alive): none
Enter Gender (m/f): f
Gamze Kaya added.

============================================
1-) Ask relation        | 2-) Add/Update person
3-) Get info of person  | 4-) Print the family tree
5-) Add marriage        | 6-) Terminate program
============================================
Please choose an operation: 4
---LEVEL 0---
Murat Aslan-Sedanur Aslan
---LEVEL 1---
Busenaz Aslan-Faruk Kaya
Mahmut Aslan-Demet Soylu
Ahmet Aslan-Mehtap Duman
Fatma Aslan
---LEVEL 2---
Gamze Kaya
```

5. Invalid Marriage

```
==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 1

--- Adding new family member ---
Please type the father name and surname: Ahmet Aslan
please type the mother name and surname: Mehtap Duman
Enter Child Name and Surname: Yunus Aslan
Enter Birthdate: 2007
Enter Death date ('none' if alive): none
Enter Gender (m/f): m
Yunus Aslan added.

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 5
Enter Name and Surname of the first person: Fatma Aslan
Enter Name and Surname of the second person: Yunus Aslan
Error: Hala-Yegen invalid marriage! (Fatma Aslan - Yunus Aslan)

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 2
1-Update Birth Year / 2-Update Death Year / 0-Cancel: 1
Enter Name and Surname of the person: Yunus Aslan
Enter Year: 2020
Yunus Aslan Year of birth updated.
```

```
==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 5
Enter Name and Surname of the first person: Yunus Aslan
Enter Name and Surname of the second person: Gamze Kaya
Error: Marriage under age 18 not valid! Yunus Aslan or Gamze Kaya under age 18.

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 2
1-) Add Person | 2-) Update Person
Please choose an operation: 2
1-Update Birth Year / 2-Update Death Year / 0-Cancel: 1
Enter Name and Surname of the person: Yunus Aslan
Enter Year: 1985
Yunus Aslan Year of birth updated.

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 5
Enter Name and Surname of the first person: Gamze Kaya
Enter Name and Surname of the second person: Yunus Aslan
success: Gamze Kaya and Yunus Aslan married.
```

6. Ask Relation

```
==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 4
---LEVEL 0---
Murat Aslan-Sedanur Aslan
---LEVEL 1---
Busenaz Aslan-Faruk Kaya
Ahmet Aslan-Mehtap Duman
Fatma Aslan
Mahmut Aslan-Demet Soylu
---LEVEL 2---
Gamze Kaya-Yunus Aslan

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 1
1. Please type name and surname of the first person: Yunus Aslan
2. Please type name and surname of the second person: Sedanur Aslan
> Torun

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 1
1. Please type name and surname of the first person: Murat Aslan
2. Please type name and surname of the second person: Mehtap Duman
> Kayinpeder

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 1
1. Please type name and surname of the first person: Mehtap Duman
2. Please type name and surname of the second person: Busenaz Aslan
> Yenge
```

```
==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 1
1. Please type name and surname of the first person: Faruk Kaya
2. Please type name and surname of the second person: Mahmut Aslan
> Eniste

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 1
1. Please type name and surname of the first person: Gamze Kaya
2. Please type name and surname of the second person: Fatma Aslan
> Yegen

==============================================
1-) Ask relation         | 2-) Add/Update person
3-) Get info of person   | 4-) Print the family tree
5-) Add marriage         | 6-) Terminate program
==============================================
Please choose an operation: 6
Terminating!
```